

41237 - Dynamics of Structures: Theory and Analysis

Report 3: Damping of an Offshore Wind Turbine

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Introduction

The present Report 3 concerns the damping of the Offshore Wind Turbine (OWT) from [2], when excited both harmonically and by a simulated double ship impact. To improve the numerical model's efficiency, system reduction is initially implemented. Then, both a viscous damper and a tuned mass damper are installed to introduce damping to the second fore-aft bending mode. Finally, direct time integration by the Newmark method is applied to analyze the structural response and assess the damper performance. Details about the nodal geometry of the OWT can be found in the previous exercise text [2].

1 Reduced multi-degree-of-freedom system

The dofs of the numerical FE model for the 3D beam structure in Fig. 1 are collected in the column vector $\mathbf{q}(t)$, while the external loading is represented by the corresponding vector function $\mathbf{f}(t)$. The dynamic equation of motion for the multi-degree-of-freedom (mdof) model can then be expressed as

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{C}\dot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{f}(t) \quad (1)$$

where $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are the mass, damping and stiffness matrix, respectively. This FE model is analyzed by modal analysis in Report 2 [2].

As damping analyses are often computationally expensive, it is often necessary to minimize the size of the FE model by some form of system reduction [1]. In the following, the number of dofs in the model are reduced by only considering the dofs in the tower and rotor as dynamic, as shown in Fig. 1. Thus, monopile and drivetrain are entirely eliminated by static condensation. Furthermore, some nodes in the rotor can be eliminated as well to reduce the computational effort. The dynamic index vector in the python-file `vib3d_owt_sysred` is therefore given as

$$\mathbf{i}_d = [1, 2, \dots, 270, 271, \dots, 276, \mathbf{i}_{d,rotor}]$$

For a fully represented rotor model with 207 nodes, the dynamic index vector for the rotor is $\mathbf{i}_{d,rotor} = [277, \dots, 1519]$. The remaining dofs, not considered as dynamic, are listed in the associated static index vector $\mathbf{i}_s$.

To obtain an efficient numerical model, the number of dynamic rotor dofs in $\mathbf{i}_{d,rotor}$ could be reduced compared to the full implementation. However, since much of the vibratory mass in the second fore-aft bending mode is associated with rotor deflections, a suitable compromise could be obtained in the system reduction between model accuracy and computational efficiency.

The partitioned stiffness matrices are extracted from the full-size matrix as:

$$\mathbf{K}_{dd} = \mathbf{K}(\mathbf{i}_d, \mathbf{i}_d) , \quad \mathbf{K}_{ds} = \mathbf{K}(\mathbf{i}_d, \mathbf{i}_s) , \quad \mathbf{K}_{sd} = \mathbf{K}(\mathbf{i}_s, \mathbf{i}_d) , \quad \mathbf{K}_{ss} = \mathbf{K}(\mathbf{i}_s, \mathbf{i}_s) \quad (2)$$

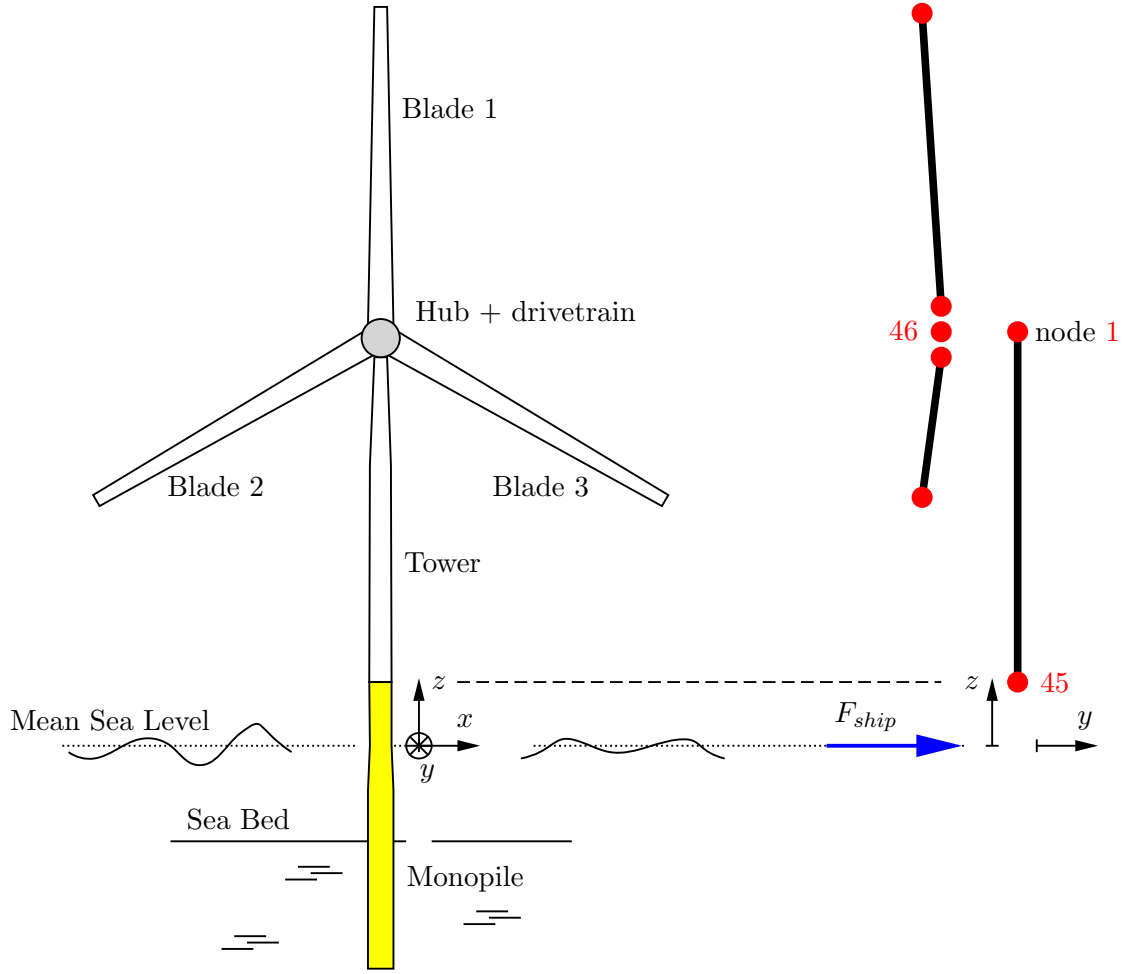


Figure 1: Geometry of the system-reduced OWT model: Monopile and drivetrain entirely eliminated by static condensation, while rotor nodes can additionally be eliminated.

and similarly for the mass matrices $\mathbf{M}_{dd}$ to $\mathbf{M}_{ss}$, while the static and dynamic parts of the load vector are $\mathbf{f}_d = \mathbf{f}(\mathbf{i}_d)$ and $\mathbf{f}_s = \mathbf{f}(\mathbf{i}_s)$, respectively. For the reduced-order model, the stiffness matrix $\bar{\mathbf{K}}$, mass matrix $\bar{\mathbf{M}}$ and load vector $\bar{\mathbf{f}}(t)$ can then be obtained by the matrix expressions provided in the course notes [1] or slides. It is noted that the ship load F_{ship} in Fig. 1 acts on an eliminated node in the monopile, thus contained in $\mathbf{f}_s$.

The reduced size damping matrix $\bar{\mathbf{C}}$ may also be obtained by the dynamic-static system reduction formula in [1]. However, in the present case, it is simply modelled by Rayleigh-damping directly in the reduced-order domain as

$$\bar{\mathbf{C}} = \alpha \bar{\mathbf{M}} + \beta \bar{\mathbf{K}} \quad (3)$$

where α and β are calibrated to a desired damping level. The reduced-order system of equations can then be written as

$$\bar{\mathbf{M}}\ddot{\mathbf{q}}_d(t) + \bar{\mathbf{C}}\dot{\mathbf{q}}_d(t) + \bar{\mathbf{K}}\mathbf{q}_d(t) = \bar{\mathbf{f}}(t) \quad (4)$$

The total number of dofs in the reduced model depends on the degree of system reduction applied to the OWT rotor.

The system reduction is already implemented in the modified `vib3d_owt_sysred` file. For the reduced model, the governing parameters are saved in the `vibprop_sysred` file, so that subsequent python packages (for Newmark time integration) can use the reduced model with a minimum of adjustments.

Exercise 8 (part 2) considers the reduced order FE model of the OWT in `vib3d_owt_sysred`.

- (a) Choose a suitable reduction of the OWT rotor in `vib3d_owt_sysred` and determine the natural frequencies for the first and second fore-aft bending modes (ω_2 and ω_{13}). Compare the natural frequencies with those obtained for the full FE model in [2].

2 Viscous damper system

For the OWT, a viscous damper system is installed vertically inside the tower, between nodes 19 and 23, as indicated in Fig. 2. The viscous damper system has a resulting viscous coefficient c , which must be tuned to damp the second fore-aft bending mode ($j = 13$) with natural frequency ω_{13} .

The location of the viscous damper system is implemented via the connectivity vector

$$\bar{\mathbf{w}}^T = [0, \dots 0, -d, 0 \dots 0, d, 0 \dots 0] \quad (5)$$

with all zero entries, except for the rotational dofs 112 and 136 at nodes 19 and 23, respectively. The apparent diameter $d = 8$ m represents an effective lever arm for the damper system. The reduced order equations of motion with the local viscous damper system installed can then be written as

$$\bar{\mathbf{M}}\ddot{\mathbf{q}}_d(t) + (\bar{\mathbf{C}} + c\bar{\mathbf{w}}\bar{\mathbf{w}}^T)\dot{\mathbf{q}}_d(t) + \bar{\mathbf{K}}\mathbf{q}_d(t) = \bar{\mathbf{f}}(t) \quad (6)$$

in which a total damping matrix is simply given as the sum of the structural (Rayleigh) damping matrix $\bar{\mathbf{C}}$ and a matrix contribution $c\bar{\mathbf{w}}\bar{\mathbf{w}}^T$ from the local viscous damper system. This extended equation of motion (6) must be implemented in Exercise 11.

An important task in the tuning of a viscous damper on a flexible structure is the determination of the so-called *locked* natural frequency $\bar{\omega}_{\infty,j}$ for the targeted vibration mode j . The subscript ∞ is chosen because it indicates that this limit corresponds to a rigid damper with $c \rightarrow \infty$ in (6). An alternative approach, which often works in practice, simply adds a very large stiffness k_∞ at or across the dofs where the damper is attached. This imposes the desired locking effect and can be implemented via the modified stiffness matrix

$$\bar{\mathbf{K}}_\infty = \bar{\mathbf{K}} + k_\infty \bar{\mathbf{w}}\bar{\mathbf{w}}^T \quad (7)$$

The locked natural frequency $\bar{\omega}_{\infty,j}$ can then be obtained by simply solving the modified undamped eigenvalue problem

$$(\bar{\mathbf{K}}_\infty - \bar{\omega}_{\infty,j}^2 \bar{\mathbf{M}})\bar{\mathbf{u}}_{\infty,j} = \mathbf{0} \quad (8)$$

The large stiffness k_∞ should be chosen sufficiently large to avoid significant deformations across the damper location, but not so large that it creates numerical (ill-posed) problems. An appropriate value of k_∞ might be between 10^3 and 10^6 times the largest diagonal element in the original stiffness matrix $\bar{\mathbf{K}}$.

Exercise 9 considers the optimal tuning of the viscous damper to the second fore-aft bending mode ($j = 13$) of the OWT model. The tuning procedure is conveniently implemented in the `vib3d_owt_sysred`-file.

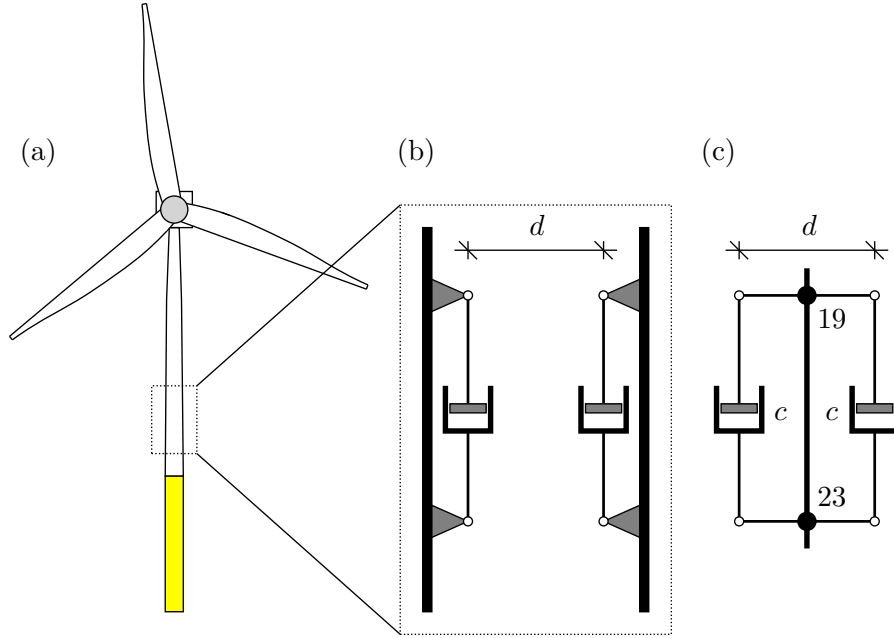


Figure 2: A set of identical viscous dampers, acting on the change in rotation θ_x between nodes 19 and 23. (a) Damper installed inside tower, (b) attached axisymmetrically along the tower wall, (c) and acting on beam element rotations via lever arm $d = 8$ m. The depicted damper configuration is assumed to be installed in the tower so that it operates identically for all rotor orientations.

- (a) Compute the locked natural frequency $\bar{\omega}_{\infty,13}$ for mode 13 (second fore-aft bending) by solving the modified eigenvalue problem in (8).
- (b) Normalize the undamped mode shape $\bar{\mathbf{u}}_{13}$ to unit modal mass and determine the modal amplitude $u_{13} = \bar{\mathbf{w}}^T \bar{\mathbf{u}}_{13}$ across the damper connection.
- (c) Determine the optimal viscous coefficient c_{opt} for damping of mode $j = 13$, the associated modal damping ratio ζ^{max} and the coupling coefficient κ . Save $\bar{\mathbf{w}}$, c_{opt} , and other governing parameters in a file `vibprop-sysred-vis`.

3 Tuned mass damper

For large-scale structures, the most common vibration mitigation strategy is the installation of Tuned Mass Dampers (TMDs). For OWTs, TMDs are often implemented as pendulum absorbers inside the tower. However, in the present case, the TMD is considered in its classic mass-spring-dashpot configuration. As shown in Fig. 3, a TMD is placed at node 26 of the OWT model, thus acting horizontally in the fore-aft direction at dof 152, with mass m , spring stiffness k and viscous dashpot coefficient c . For the system-reduced model, the connectivity vector can thereby be expressed as

$$\bar{\mathbf{w}}^T = [0, \dots, 0, 1, 0, \dots, 0] \quad (9)$$

with the single unit value at the horizontal dof 152. The displacement of the TMD is in Fig. 3 denoted as u_d .

- (a) Damper installed inside tower, (b) attached axisymmetrically along the tower wall, (c) and acting on beam element rotations via lever arm $d = 8$ m.

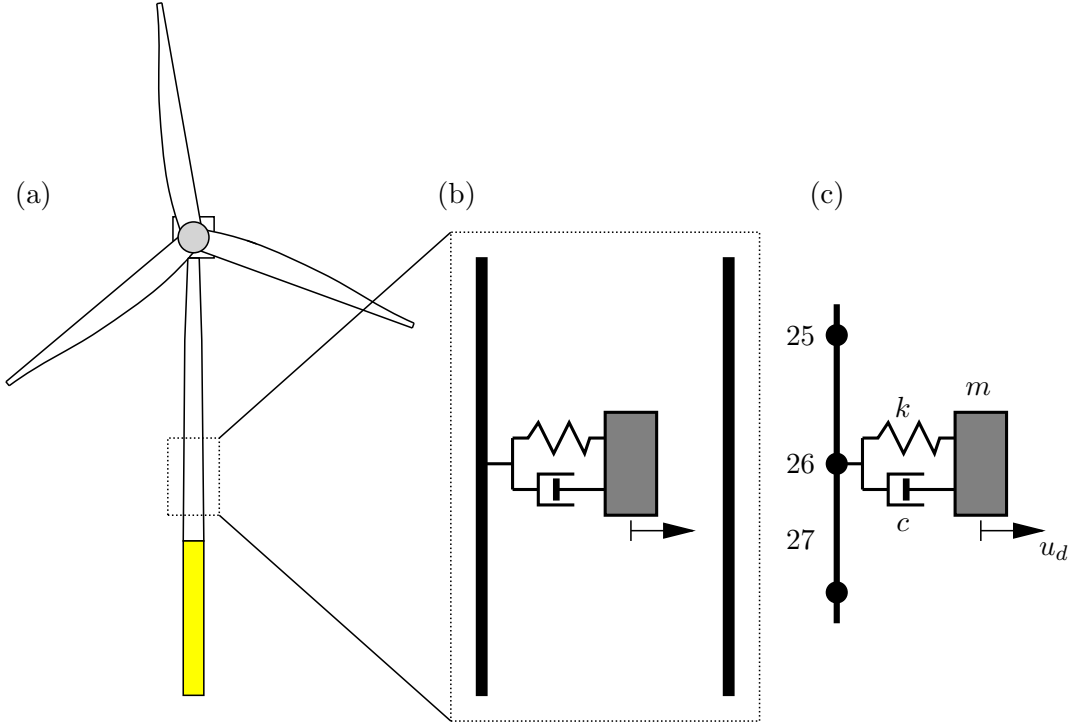


Figure 3: (a,b) TMD acts horizontally inside tower, (c) acting on the horizontal displacement dof 152 at node 26 in the fore-aft direction. The TMD parameters are mass m , stiffness k and viscous coefficient c , while u_d represents the absolute displacement of the TMD mass.

3.1 MDOF structure with TMD

The force exerted on the structure and conversely on the TMD mass is defined by the deformation of the spring-dashpot element,

$$f_d(t) = c(\dot{u}_d - \bar{\mathbf{w}}^T \dot{\mathbf{q}}_d) + k(u_d - \bar{\mathbf{w}}^T \mathbf{q}_d) \quad (10)$$

The set of governing equations for the flexible structure and TMD can then be written as

$$\begin{aligned} \bar{\mathbf{M}}\ddot{\mathbf{q}}_d(t) + \bar{\mathbf{C}}\dot{\mathbf{q}}_d(t) + \bar{\mathbf{K}}\mathbf{q}_d(t) - \bar{\mathbf{w}}f_d(t) &= \bar{\mathbf{f}}(t) \\ m\ddot{u}_d(t) + f_d(t) &= 0 \end{aligned} \quad (11)$$

The damper force f_d is now eliminated by substitution of (10) into (11), which can then be written in the compact matrix form

$$\tilde{\mathbf{M}}\ddot{\tilde{\mathbf{q}}}(t) + \tilde{\mathbf{C}}\dot{\tilde{\mathbf{q}}}(t) + \tilde{\mathbf{K}}\tilde{\mathbf{q}}(t) = \tilde{\mathbf{f}}(t) \quad (12)$$

for the extended displacement vector

$$\tilde{\mathbf{q}}(t) = \begin{bmatrix} \mathbf{q}_d(t) \\ u_d(t) \end{bmatrix} \quad (13)$$

and the augmented system matrices

$$\tilde{\mathbf{M}} = \begin{bmatrix} \bar{\mathbf{M}} & \mathbf{0} \\ \mathbf{0}^T & m \end{bmatrix}, \quad \tilde{\mathbf{C}} = \begin{bmatrix} \bar{\mathbf{C}} + c\bar{\mathbf{w}}\bar{\mathbf{w}}^T & -c\bar{\mathbf{w}} \\ -c\bar{\mathbf{w}}^T & c \end{bmatrix}, \quad \tilde{\mathbf{K}} = \begin{bmatrix} \bar{\mathbf{K}} + k\bar{\mathbf{w}}\bar{\mathbf{w}}^T & -k\bar{\mathbf{w}} \\ -k\bar{\mathbf{w}}^T & k \end{bmatrix} \quad (14)$$

The external load vector

$$\tilde{\mathbf{f}}(t) = \begin{bmatrix} \bar{\mathbf{f}}(t) \\ 0 \end{bmatrix} \quad (15)$$

assumes that no external load is applied to the TMD, as it is placed inside the structure. The augmented system of equations in (11) is implemented in Exercise 11.

3.2 Reduction to modal SDOF structure with TMD

The TMD is calibrated to the dominant vibration mode $j = 13$. For the TMD tuning and design, the structural displacement is therefore approximated by only a single mode as

$$\mathbf{q}_d(t) = \bar{\mathbf{u}}_{13} r_{13}(t) \quad , \quad \bar{\mathbf{w}}^T \bar{\mathbf{u}}_{13} = 1 \text{ (normalization)} \quad (16)$$

This representation is substituted into (11) after f_d has been eliminated by (10). When the first equation in (11) is pre-multiplied with $\bar{\mathbf{u}}_{13}^T$, the system of equations is reduced to the 2-dof system of equations that is used to calibrate the TMD parameters. As the mode shape $\bar{\mathbf{u}}_{13}$ is normalized to unit deflection at the location of the TMD ($\bar{\mathbf{w}}^T \bar{\mathbf{u}}_{13} = 1$), the reduced system of modal equations can be written as

$$\begin{aligned} m_{13} \ddot{r}_{13}(t) + (c_{13} + c) \dot{r}_{13}(t) + (k_{13} + k) r_{13}(t) - \dot{c} u_d(t) - k u_d(t) &= f_{13}(t) \\ m \ddot{u}_d(t) + \dot{c} u_d(t) + k u_d(t) - \dot{c} r_{13}(t) - k r_{13}(t) &= 0 \end{aligned} \quad (17)$$

where modal mass, damping, stiffness and load are defined as

$$m_{13} = \bar{\mathbf{u}}_{13}^T \bar{\mathbf{M}} \bar{\mathbf{u}}_{13} \quad , \quad c_{13} = \bar{\mathbf{u}}_{13}^T \bar{\mathbf{C}} \bar{\mathbf{u}}_{13} \quad , \quad k_{13} = \bar{\mathbf{u}}_{13}^T \bar{\mathbf{K}} \bar{\mathbf{u}}_{13} \quad , \quad f_{13}(t) = \bar{\mathbf{u}}_{13}^T \bar{\mathbf{f}}(t) \quad (18)$$

In the case without structural damping ($c_{13} = 0$), the pair of equations (17) recovers the governing equations for the single-mass structure with a TMD in [1] when $x_d = u_d - \bar{\mathbf{w}}^T \mathbf{q}_d = u_d - u_{13}$ represents the relative TMD displacement for the single-mode approximation introduced in (16). The derivations in this subsection are included to demonstrate that the tuning expressions in [1] can be applied to the present OWT FE model.

3.3 Tuned mass damper calibration

The theory for the TMD enables a determination of the mass ratio μ based on a desired level of modal damping ζ_{des} . This relation can be written as

$$\zeta_{des} = \frac{1}{2} \zeta_d = \sqrt{\frac{1}{8} \frac{\mu}{1 + \mu}} \quad (19)$$

which is initially used in the present exercise to choose the TMD mass m . Subsequently, the purpose of the present exercise is to optimally calibrate the remaining TMD parameters (k and c) to the second fore-aft vibration mode $j = 13$ with natural frequency $\bar{\omega}_{13}$ representing ω_0 in the course notes [1]. As for the tuning of the local viscous damper in Exercise 9, the implementation of the TMD calibration is conveniently conducted directly in the `vib3d_owt_sysred`-file, which contains the reduced-order matrices.

Exercise 10 *conducts the optimal TMD tuning to the second fore-aft bending mode ($j = 13$) of the OWT.*

- (a) Use the design formula (19) to determine the TMD mass ratio for a desired damping ratio $\zeta_{des} = 0.02$. Normalize the mode shape $\bar{\mathbf{u}}_{13}$ so that $\bar{\mathbf{w}}^T \bar{\mathbf{u}}_{13} = 1$, calculate the modal mass m_{13} and determine the TMD mass $m = \mu m_{13}$ with respect to the second fore-aft bending mode.
- (b) Use the calibration formulae for the TMD in [1] to determine the optimal angular frequency ω_d and damping ratio ζ_d . Then obtain the corresponding values for the TMD stiffness k and viscous coefficient c . Save $\bar{\mathbf{w}}$, m , c and k , and other governing parameters in a file `vibprop_sysred_tmd`.

4 Direct Newmark time integration

When installing local damping devices on the structure, the orthogonality conditions associated with modal analysis are violated. This requires that structural damping is represented by a damping matrix, such as the Rayleigh damping matrix in (3), and that the governing equations of motion in (6) or (12) are solved by direct time integration, presently using the Newmark method in the `timeintframe`-package (download from DTU Learn).

Exercise 11 uses the Newmark method in `timeintframe` to assess the ability of the viscous damper and the TMD to mitigate both the resonant and double-impact loading at MSL location.

- (a) Implement the Rayleigh damping matrix in (3) so that $\zeta_2 = \zeta_{13} = 0.025$ for the first and second fore-aft bending modes.
- (b) Implement the ship load F_{ship} from [2] in the system-reduced model to obtain the load vector $\bar{\mathbf{f}}(t)$. Do so for both the harmonic loading and the ship impact loading with $\omega_f = \bar{\omega}_{13}$.
- (c) Use the Newmark method to time-integrate the equation of motion (6) without damper ($c = 0$). Plot relevant time response histories and compare the response with that obtained by modal analysis in Report 2. Do so for both harmonic and double-impact loading.
- (d) Use the Newmark method to time-integrate the equation of motion (6) for the tuned viscous damper system in `vibprop_sysred_vis`. Plot relevant time response histories and comment on the performance of the viscous damper system with respect to both harmonic and double-impact loading.
- (e) Implement the ship loading from (b), the system matrices and the TMD parameters from `vibprop_sysred_tmd` into the augmented system of equations (12) for the combined OWT-TMD system. Time-integrate these coupled equations by the Newmark method for both harmonic and double-impact loading. Plot relevant time response histories and comment on the TMD's ability to mitigate harmonic and impact loading. Furthermore, compare the tmd damping performance with that of the viscous damper system in (d).

Report instructions

This Report 3 is submitted by groups of two to three students via the DTU Learn platform. It is strongly encouraged to collaborate with other students/groups via the available learning platforms and seek help from teaching assistants and/or the lecturer.

The present report should be limited to 12 pages, comprising:

- A short introduction to the dynamic problem and a summary of the main content and results.
- A section for each of the exercises, in which the specific questions are solved and results are discussed.
- A compact final discussion.

The report layout should use reasonable font size, margin and line spacing (as in the present exercise text). Furthermore, the written content, mathematical derivations and figures should be prepared and organized to present the dynamic analysis in a compact form that will allow the reader to reproduce the results. Although it is not allowed to directly use figures from this or previous report exercise texts, student reports may refer to background material via the references.

The reports are assessed using several criteria. A major part of the assessment concerns the content and results presented in the report, and the ability to discuss results and the underlying physics. Furthermore, the reports are assessed with respect to layout, (English) grammar and style in general.

References

- [1] S. Krenk. *Dynamics of Structures*, Department of Mechanical Engineering, Technical University of Denmark, pp. 173, 2013.
- [2] P. Elbek, J.D. Richardt & J. Høgsberg. 41237 - Dynamics of Structures: Theory and Analysis. Report 2: The Dynamics of an Offshore Wind Turbine, *Technical university of Denmark*, Course 41237, 2026.